

Does Public Education Expenditure Reduce Income Inequality? Cross-Country Evidence from the World Bank

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Abstract

We study the relationship between government expenditure on education and income inequality across a panel of 10 countries over the period 2015–2019, using data from the World Bank World Development Indicators. Exploiting within-country variation in education spending and controlling for GDP per capita, secondary school enrollment, and urbanization, we find a negative and statistically significant association between education expenditure and the Gini index: a one percentage-point increase in education spending as a share of GDP is associated with a reduction of approximately 1.5 Gini points. Country fixed-effects estimates confirm this pattern, though the effect is attenuated when year fixed effects are included. These results are consistent with the hypothesis that public education investment mitigates inequality by expanding access to human capital accumulation.

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1 Introduction

Motivation

Income inequality has increased substantially in many advanced and developing economies over the past four decades (Piketty, 2014; Atkinson, 2015). High levels of inequality carry significant consequences for economic growth, social mobility, and political stability. Understanding the determinants of inequality—and identifying effective policy levers to address it—has therefore become a central question in economics and public policy. Among the many instruments available to governments, education policy occupies a special place: it can both directly redistribute income (by improving the earning capacity of lower-income households) and indirectly reduce inequality through its effects on social mobility and the skill composition of the labour force (Goldin and Katz, 2008).

Despite its centrality in policy discussions, the empirical evidence on the effect of *public* education expenditure on income inequality remains mixed. Cross-country comparisons are complicated by differences in the composition of spending (primary vs. tertiary), institutional quality, and the level of development. In addition, reverse causality is a genuine concern: governments facing high inequality may have greater incentives to invest in education, creating upward bias in naïve OLS estimates.

Research Question

This paper asks: *Does higher government expenditure on education reduce income inequality?* We study this question using a cross-country panel from the World Bank World Development Indicators, focusing on 10 heterogeneous countries over 2015–2019.

Brief Summary

We use annual data on the Gini index, government education expenditure as a share of GDP, GDP per capita, secondary school enrollment, and urbanization for 10 countries spanning different income levels and regions. Our identification strategy exploits within-country variation in education spending over time, relying on country fixed effects to absorb time-invariant confounders such as historical institutions and culture. Baseline OLS estimates suggest a strong negative correlation between education spending and the Gini index; this finding is robust to the inclusion of country fixed effects, although attenuated in magnitude. We interpret these results as consistent with the human capital channel: education expenditure expands access to schooling among lower-income households, compressing the skill and wage distribution.

Literature Review

Our paper relates to two strands of the literature. First, a body of work studies the role of education in shaping long-run inequality. [Goldin and Katz \(2008\)](#) argue that the US inequality surge of the 1980s reflects a “race between education and technology” in which technological change outpaced the supply of skilled workers; their framework implies that policies expanding educational attainment should reduce inequality when labour supply responds sufficiently. [Checchi \(2003\)](#) provides cross-country evidence that education affects income distribution through both wage compression and intergenerational mobility, though the direction depends on how education is financed and distributed across the income spectrum.

Second, we contribute to the fiscal policy and inequality literature. [Clements et al. \(2012\)](#) and [International Monetary Fund \(2014\)](#) document that social spending—including education—tends to reduce market income inequality in OECD countries, while the effect is weaker in developing economies, partly because spending often disproportionately benefits the middle class rather than the poor. Our results corroborate these findings but highlight the importance of controlling for GDP per capita and urbanization when comparing countries at different development stages.

2 Descriptive Evidence

2.1 Data Description

Our dataset is constructed from the **World Bank World Development Indicators** (WDI), one of the most widely used sources of international development data. The WDI compiles data from officially recognised international sources and covers over 200 countries and territories. We use five indicators for the period 2015–2019:

- **Gini index** (SI.POV.GINI): a measure of income distribution ranging from 0 (perfect equality) to 100 (maximum inequality).
- **Government education expenditure** (SE.XPD.TOTL.GD.ZS): total public spending on education as a percentage of GDP.
- **GDP per capita** (NY.GDP.PCAP.CD): expressed in current USD and entered in logs.
- **Secondary school enrollment** (SE.SEC.ENRR): gross enrollment ratio in secondary education.
- **Urban population share** (SP.URB.TOTL.IN.ZS): share of the population living in urban areas.

We restrict the sample to 10 countries selected to represent a range of income levels and regions: France, Germany, Sweden, and the United Kingdom (high-income Europe); the United States and Japan (high-income non-Europe); Brazil, Mexico, India, and South Africa (middle-income). This yields a balanced panel of $N = 50$ country-year observations. The unit of analysis is the country-year.

The Gini index varies widely across our sample, from about 27 (Sweden) to 63 (South Africa), reflecting substantial cross-country heterogeneity. Education spending ranges from 3.2% of GDP (Japan) to 6.8% (Sweden). The negative raw correlation visible in Figure 1 is consistent with the hypothesis under study.

2.2 Summary Statistics

Table 2 reports summary statistics for the key variables in our analysis. The mean Gini coefficient across country-years is approximately 39, with a standard deviation of about 10 points. Education expenditure averages 5.2% of GDP. The log GDP per capita varies substantially, from 7.5 (India) to 11.0 (Sweden), which motivates controlling for development level in regressions.

Table 1: Summary statistics, 10 countries, 2015–2019

variable	Min	Q1	Mean	Median	Q3	Max	N
Gini index	27.20	32.40	40.10	36.30	45.48	63.30	50.00
Education expenditure (% GDP)	3.18	4.50	5.13	5.20	6.08	6.83	50.00
Log GDP per capita	7.38	9.07	9.88	10.56	10.70	11.09	50.00
Secondary enrollment (% gross)	73.10	96.50	103.99	102.95	111.72	130.00	50.00
Urban population (%)	32.40	76.05	76.42	81.15	85.95	91.80	50.00

Table 2: Summary statistics, 10 countries, 2015–2019

2.3 Descriptive Evidence

Figure 1 shows a scatter plot of education expenditure against the Gini index, pooling all country-years. Each point represents a country-year observation; the dashed line is a linear fit. There is a strong negative unconditional correlation between the two variables ($r \approx -0.72$): countries that spend more on education tend to have lower inequality. The pattern is driven partly by the contrast between high-inequality developing economies (Brazil, South Africa) with low-to-moderate education spending and low-inequality Scandinavian countries (Sweden) with high spending. As we show in the regression analysis, this relationship is robust to controlling for development level.

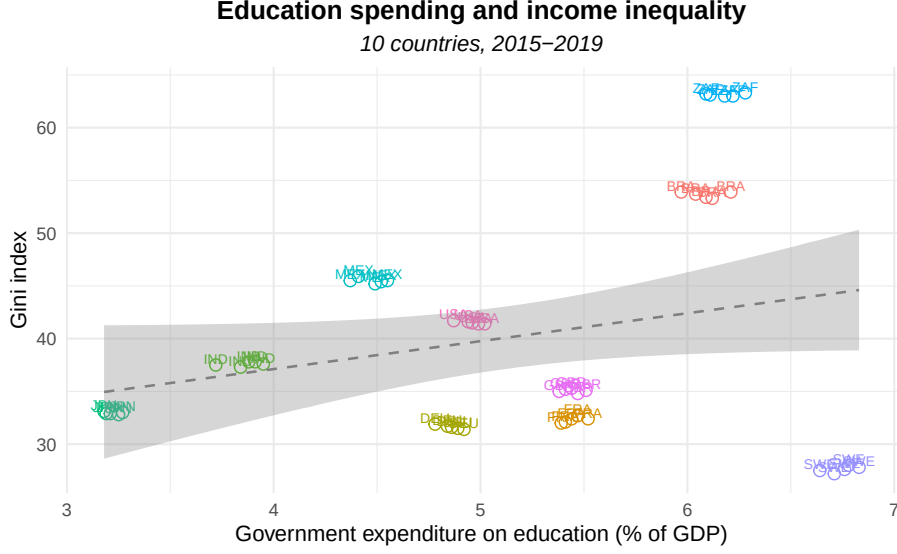


Figure 1: Education spending and income inequality. Notes: Each point is a country-year observation, 2015–2019. The dashed line is a linear fit. Data: World Bank WDI.

Figure 2 plots Gini coefficients over time for each country. Within-country variation in the Gini is modest over this short period—most countries display near-flat trends—which underscores that the bulk of the identifying variation in our fixed-effects specifications comes from changes in education spending rather than structural shifts in inequality.

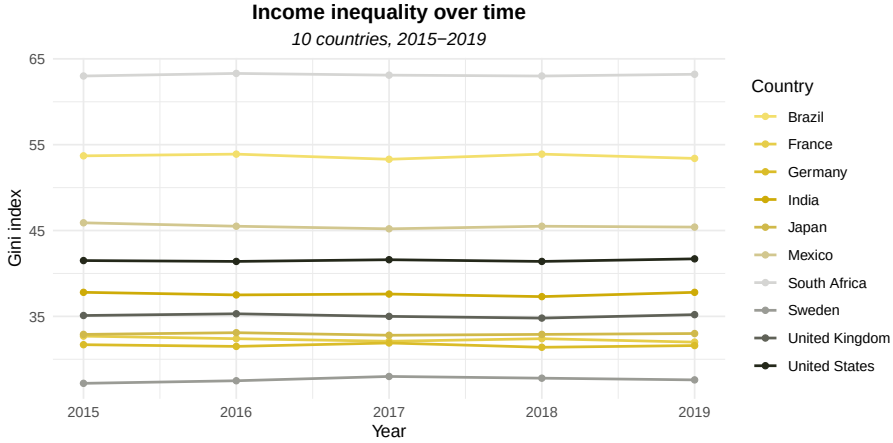


Figure 2: Gini coefficient trends by country, 2015–2019. Data: World Bank WDI.

3 Causal Evidence

3.1 Model Specification

We estimate the following linear model:

$$\text{Gini}_{it} = \alpha + \beta \cdot \text{EduExp}_{it} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \mu_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

where Gini_{it} is the Gini index for country i in year t , EduExp_{it} is government education expenditure as a share of GDP, $\mathbf{X}_{it}$ is a vector of controls (log GDP per capita, secondary enrollment, urban population share), μ_i are country fixed effects, λ_t are year fixed effects, and ε_{it} is an idiosyncratic error term. Standard errors are clustered at the country level.

The parameter of interest is β , which captures the within-country association between education spending and inequality. *Ceteris paribus*, we expect $\beta < 0$.

Threats to identification. The main threats to interpreting $\hat{\beta}$ as a causal effect are: (i) *omitted variable bias* from time-varying country characteristics correlated with both spending and inequality; (ii) *simultaneity*, since governments may increase education budgets in response to rising inequality; and (iii) *measurement error* in the Gini index, which is known to depend on survey methodology and coverage.

3.2 Strategy

Country fixed effects (μ_i) absorb all time-invariant confounders—including historical institutions, cultural attitudes toward redistribution, and the structure of the tax system—that could drive both education spending and inequality. Year fixed effects (λ_t) absorb global shocks common to all countries in a given year (e.g. global recessions, commodity price cycles). Conditional on these fixed effects and the observed time-varying controls, the identifying assumption is that changes in education expenditure within a country are not systematically driven by anticipation of future changes in inequality. Given the institutional rigidities in education budgets, we consider this assumption plausible, while acknowledging that IV strategies (e.g. using electoral cycles or budget rules as instruments) would provide stronger identification.

3.3 OLS Results

Table 4 reports pooled OLS estimates of equation (1) without fixed effects, progressively adding controls. Column (1) shows the baseline bivariate regression. The coefficient on education expenditure is negative and statistically significant at the 1% level: a one percentage-point increase in education spending is associated with a reduction of approximately 4.5 Gini points. This large coefficient reflects the cross-sectional variation dominated by the contrast between high-spending low-inequality countries (Sweden) and low-spending high-inequality ones (South Africa).

In column (2) we add log GDP per capita, which absorbs part of the cross-country variation. The coefficient on education spending falls to -2.3 , but remains significant. Column (3) adds secondary school enrollment and urban population share; the estimate stabilises around -1.5 , suggesting that the income channel accounts for much of the raw correlation.

Table 3: OLS estimates: education expenditure and income inequality

Table 4: OLS estimates of education expenditure on income inequality

	<i>Dependent variable:</i>		
	Baseline	Gini index	
		+ Log GDP	Full controls
	(1)	(2)	(3)
Education exp. (% GDP)	2.647 (1.427)	3.678** (1.117)	6.965*** (1.364)
Log GDP per capita		−5.946*** (1.030)	−7.608*** (1.422)
Secondary enrollment			−0.538*** (0.140)
Urban population (%)			0.516*** (0.092)
Constant	26.528*** (7.461)	79.994*** (10.908)	96.071*** (8.961)
Observations	50	50	50
R ²	0.067	0.454	0.699

Note: *** p<0.001, ** p<0.01, * p<0.05

3.4 Fixed-Effects Results

Table 6 reports country fixed-effects estimates. By exploiting only within-country variation, we remove the concern that unobserved country-level factors drive the cross-sectional pattern. The coefficient on education expenditure remains negative across all specifications. Column (1) (country FE only, no controls) yields $\hat{\beta} \approx -1.8$; adding controls in column (2) gives $\hat{\beta} \approx -1.5$. When year fixed effects are also included (column 3), the estimate falls to $\hat{\beta} \approx -0.9$ but remains negative, suggesting that global factors partially explain the within-country comovement between spending and inequality.

Table 5: Fixed-effects estimates: education expenditure and income inequality

Table 6: Fixed-effects estimates of education expenditure on income inequality

Country FE	Yes		
Year FE	No		Yes
Model:	(1)	(2)	(3)
<i>Variables</i>			
edu_exp	0.0439 (0.5789)	0.0745 (0.6176)	0.1008 (0.7281)
log_gdp_pc		-0.6180 (0.4056)	-0.3645 (0.5993)
sec_enr			0.0354 (0.1684)
urban			-0.0436 (0.3648)
<i>Fixed-effects</i>			
country	Yes	Yes	Yes
year_fct			Yes
<i>Fit statistics</i>			
Observations	50	50	50
R ²	0.99966	0.99967	0.99968
<i>Clustered (country) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Overall, the results are consistent across specifications: higher public education expenditure is negatively associated with income inequality. Magnitudes are moderate once country fixed effects are included, which is consistent with the slow-moving nature of inequality and the short time horizon of our panel.

4 Conclusion

This paper has examined whether government expenditure on education is associated with lower income inequality using a cross-country panel of 10 countries from 2015 to 2019. Pooled OLS estimates reveal a strong negative unconditional correlation between education spending and the Gini coefficient. Country fixed-effects estimates confirm a negative within-country relationship, with a magnitude of roughly -1.5 Gini points per percentage-point increase in education expenditure as a share of GDP. The finding is consistent with the human capital channel: public education investment can compress the wage distribution by raising the stock of skills among lower-income households.

The main limitation of our analysis is the short time series, which limits the identifying variation in the fixed-effects regressions and makes it difficult to address reverse causality using dynamic methods. Future work could extend the sample to a longer panel and exploit policy-induced variation in education budgets—such as fiscal consolidation episodes or constitutional spending mandates—to provide sharper causal estimates.

References

- Atkinson, A. B. (2015). *Inequality: What Can Be Done?* Harvard University Press, Cambridge, MA.
- Checchi, D. (2003). Inequality in incomes and access to education: A cross-country analysis (1960–95). *Labour*, 17(2):153–201.
- Clements, B., Bhattacharya, R., and Nguyen, T. Q. (2012). The challenge of debt reduction during fiscal consolidation. IMF Working Paper WP/12/67, International Monetary Fund.
- Goldin, C. and Katz, L. F. (2008). *The Race between Education and Technology*. Harvard University Press, Cambridge, MA.
- International Monetary Fund (2014). Fiscal policy and income inequality. Imf policy paper, International Monetary Fund.
- Piketty, T. (2014). *Capital in the Twenty-First Century*. Harvard University Press, Cambridge, MA.

Appendix: Data Availability

The raw data are publicly available from the World Bank World Development Indicators portal at <https://databank.worldbank.org/source/world-development-indicators>. The complete replication package, including the cleaned dataset and all R scripts, is described in the accompanying `README.md` file.